



STATE BANK OF INDIA · GLOBAL FINTECH FEST HACKATHON 2026

SBI Pulse

An agentic AI that senses **behaviours, financial patterns & life events** — and runs proactive, adaptive engagement **journeys, not nudges**.

PILLAR **03**

Digital Engagement

Theme · Agentic AI & Emerging Tech

THE PROBLEM STATEMENT WE SOLVE

Pillar 03 — Digital Engagement

"Create AI-driven engagement models that **proactively interact** with customers based on **behaviours, financial patterns, and life events.**"

— SBI GFF HACKATHON 2026 · FOCUS AREAS · PILLAR 03

Personalised engagement

Every message generated per-customer, per-situation, per-language — not a segment blast.

Intelligent financial interactions

The agent reads the transaction stream and reasons about what would genuinely help — including staying silent.

AI-led banking journeys

Multi-step conversations that branch on the customer's reply — run end-to-end by the agent.

WHY THIS PROBLEM — THE ENGAGEMENT GAP

SBI's biggest untapped asset is the under-engaged customers it already has

34%

of new accounts go **dormant within a year** — global banking-industry figure

StriveCloud, "Banking App Engagement" playbook — industry data, early 2025

2.3

avg **products per customer** at SBI — the cross-sell ceiling to break

MatrixBCG, SBI customer & target-market analysis, 2025

5–7×

cheaper to **retain** a customer than acquire one

Vantage Point, bank customer-retention strategies, 2026 (echoed by Backbase)

~100M

YONO users today; SBI targets **20 crore in 2 years**

SBI Chairman C S Setty — PTI / The Week & ZeeBiz, Dec 2025

Banks answer this with batch campaigns and one-shot nudges optimised for clicks. Result: fatigue, opt-outs, thin trust. The 2026 frontier is **anticipatory, journey-level engagement** — Gartner projects up to a 35% operational edge for journey-orchestrating banks (via CMSWire, 2026).

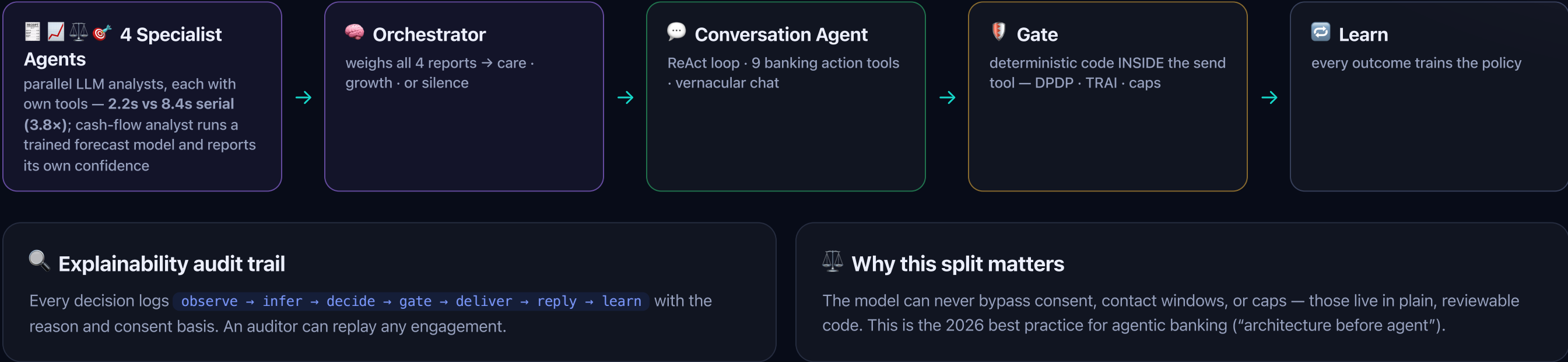
THE SOLUTION — HOW WE SOLVE IT

Stop sending nudges. **Run journeys.**

- 1 Sense.** Four specialist AI agents investigate every customer in parallel — spending, cash-flow, risk & consent, product-fit — each with its own tools.
- 2 Decide.** An orchestrator agent weighs the four reports and rules: engage with care, engage for growth — or stay silent.
- 3 Engage.** A conversation agent executes in the customer's language, adapting to every reply — and takes real banking actions on agreement.
- 4 Restrain.** On detected financial stress: all cross-sell suppressed, help first. Trust is the product.
- 5 Learn.** Every outcome trains the policy — the system gets measurably better with use.

HOW IT WORKS — THE AGENT MESH

4 parallel specialists → orchestrator → conversation agent



THE AI CORE — THREE TIERS OF AGENTS, ALL REAL

No scripts. No if/else. The models decide.

1 • Specialist analysts

Four LLM agents with distinct roles and restricted tool access investigate in parallel. The cash-flow analyst reported **“model projection is unreliable”** on its own — it read the forecaster's holdout skill.

2 • Orchestrator

Synthesizes the four reports and decides — including **“stay silent”** (Rohan: “financially healthy, no intervention required”) and consent-aware restraint (Neha, unprompted).

3 • Conversation agent

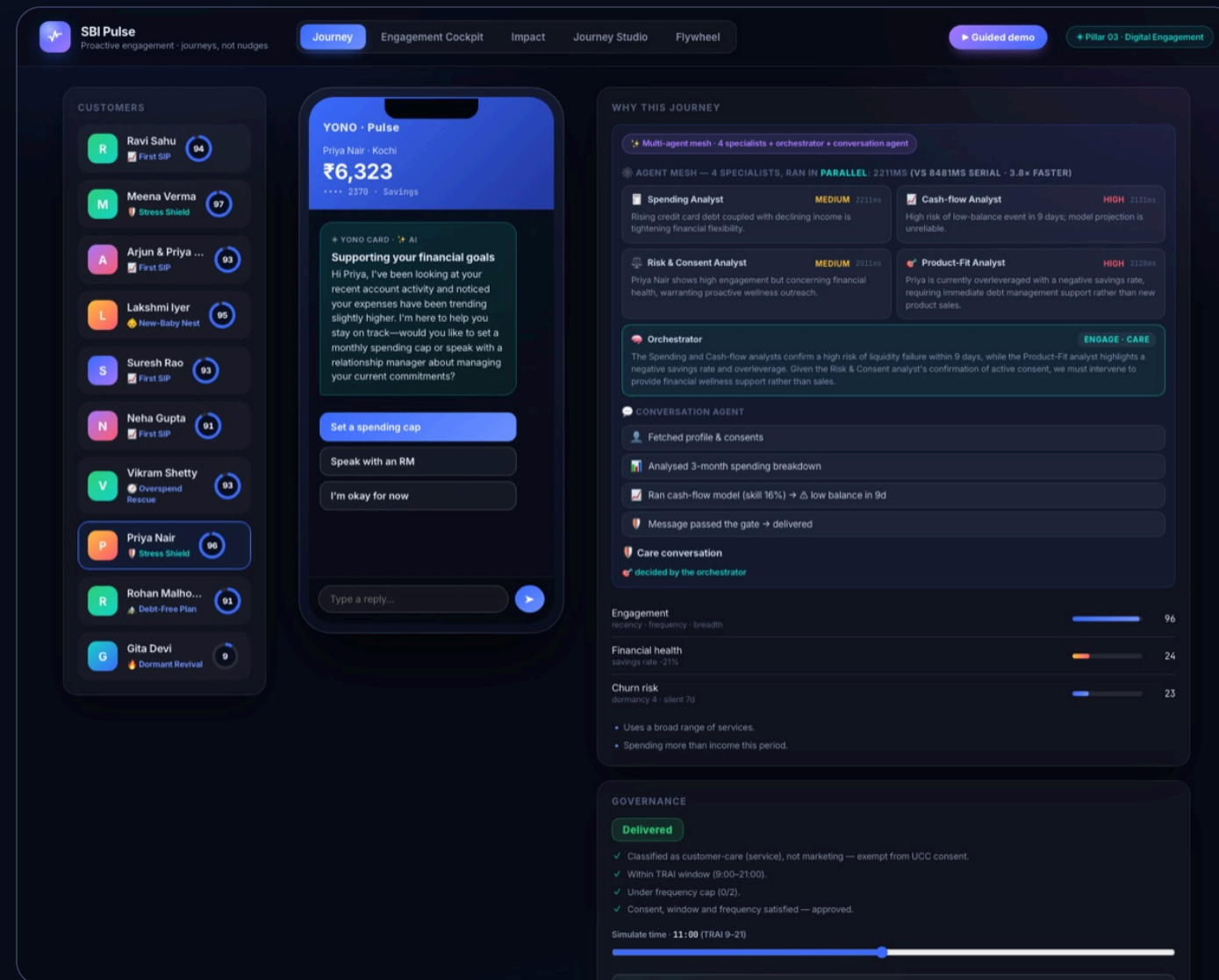
A ReAct tool-calling loop: free-text chat in **Hindi, Odia or English**, and real actions — “set a ₹15,000 cap” → it calls the banking API, sets it, confirms.

+ Trained forecaster

Per-customer cash-flow model, **validated on a 30-day holdout** (skill up to 94% vs naive — negatives shown too). Predicted Priya's low balance **9 days ahead**.

Lightweight production LLM · structured JSON + function calling · model-agnostic
(swappable to an India-hosted / Banking-BHASHINI model) · rule fallback exists for outages
and is labelled “no AI” in the UI.

One screen: the mesh runs, the orchestrator rules, **care in Hindi** ships



Priya's real (synthetic) statement → 4 specialists investigate in parallel (timings on screen) → the orchestrator rules ENGAGE-CARE → the conversation agent writes the message → the deterministic gate approves — every step on the audit trail.

WHAT'S IN THE BUILD — FEATURE MAP

Five product surfaces, **one brain**



Journey

YONO-style phone where the AI journey unfolds live — signals, reasoning trace, message, free-text chat, streak gamification.



Engagement Cockpit

Fleet view: engagement / churn / dormancy scores, journey mix, care interventions, delivered vs suppressed.



Impact

1,500-customer cohort, 80/20 treatment vs **randomised holdout** — uplift measured on outcomes, not clicks.



Journey Studio

Every journey as a governed no-code **state machine** — triggers, steps, branches, care flags. Business designs; policy enforces.



Flywheel

Live learning curve of the bandit policy climbing from random baseline toward the oracle ceiling. Run more rounds on stage.

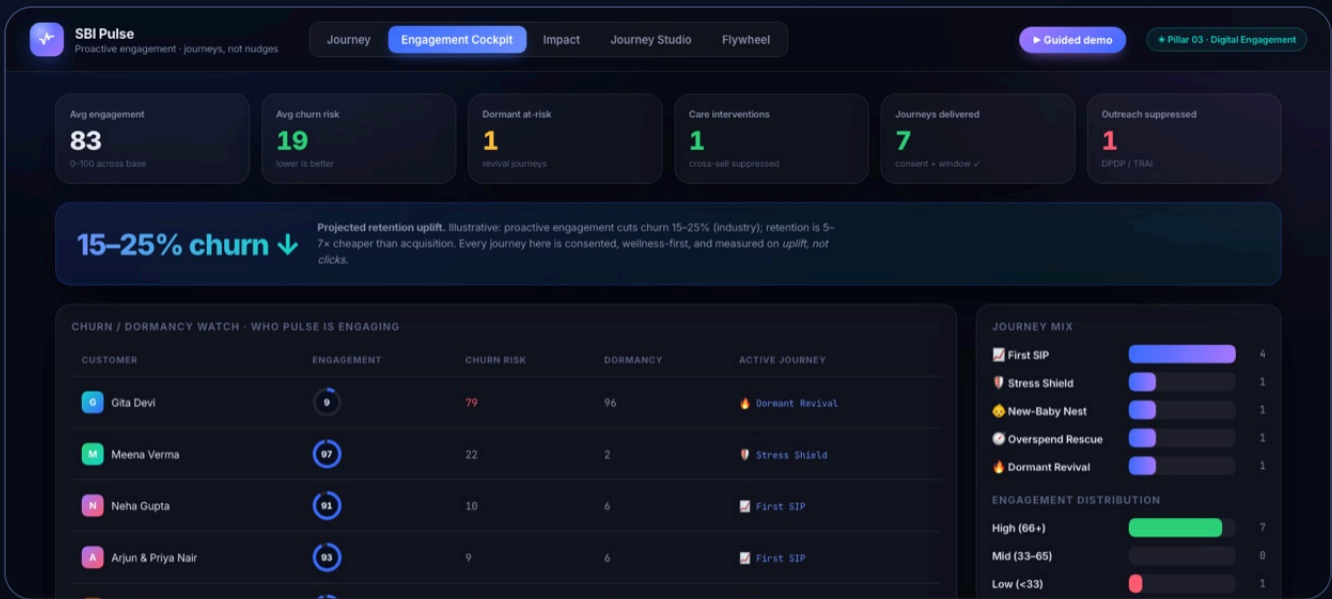


Governance layer

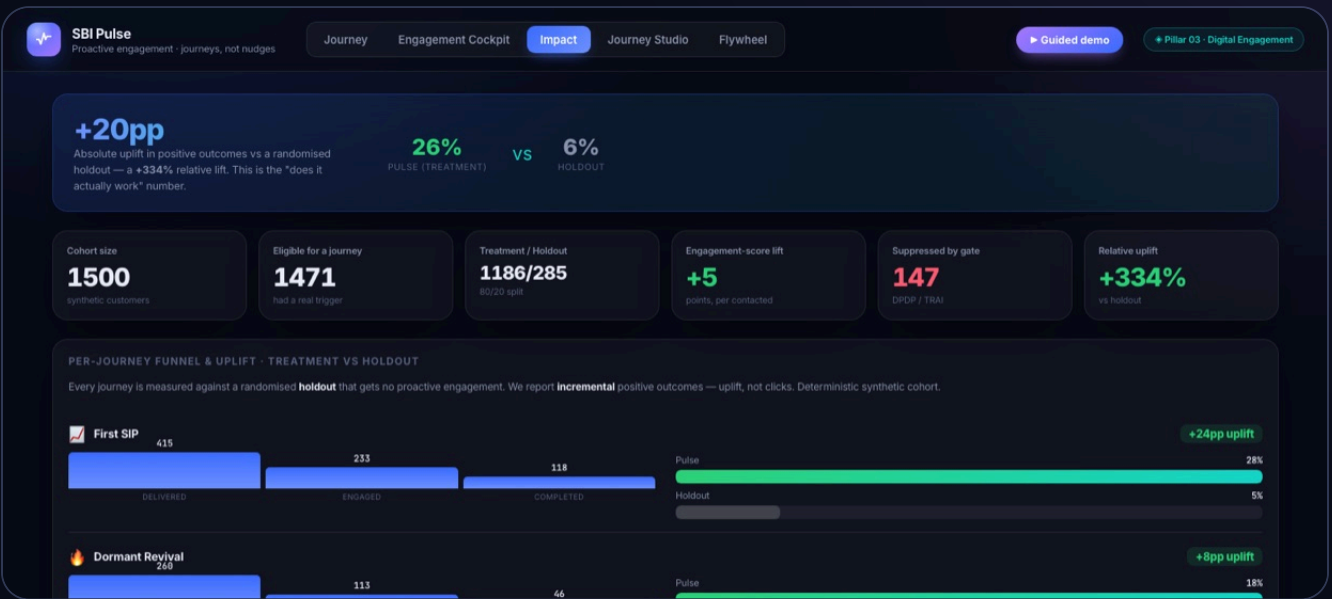
DPDP purpose-scoped consent, TRAI 9am–9pm window, frequency caps, stress suppression, human-in-loop for high value, full audit.

5 journeys (First SIP · Overspend Rescue · Dormant Revival · Stress Shield · New-Baby Nest) · 8 personas · 3 languages · guided demo mode · 12-slide pitch deck.

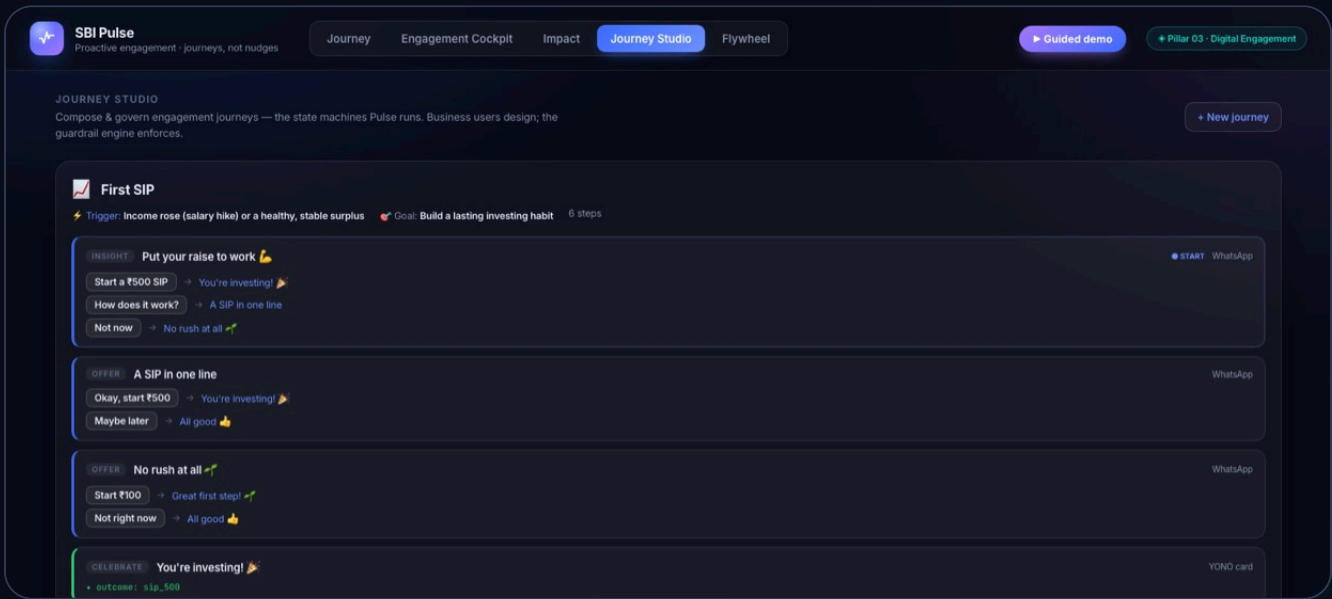
Four more surfaces, running today



🇮🇳 Engagement Cockpit — fleet scores & compliance counters



📈 Impact — +20pp uplift vs randomised holdout



COMPETITIVE ANALYSIS

Where Pulse sits vs the field

	BANK CHATBOTS (INCL. YONO Q&A)	CAMPAIGN / NUDGE ENGINES	GLOBAL LEADERS (BOFA ERICA, WELLS NBA)	SBI PULSE
Proactive (not reactive)	✗ waits to be asked	✓ batch blasts	✓	✓ per-customer agent
Multi-step, adapts to replies	~ scripted trees	✗ one-shot	~ partial	✓ LLM journeys that branch
Knows when NOT to sell	✗	✗ optimises clicks	~	✓ stress → care, sale suppressed
Vernacular generation	~ translations	✗	✗ (US-centric)	✓ Hindi · Odia · English, generated
Measured on uplift vs holdout	✗	✗ CTR	~ internal	✓ built-in, on-screen
Learns from every outcome	✗	~ A/B only	~	✓ live bandit flywheel
Consent-native + auditable	~	✗ DLT at best	~	✓ deterministic gate + full trail

EVERY TECHNOLOGY IN THE BUILD

Small, sovereign-ready, production-shaped

🌟 AI & learning

- multi-agent mesh (4 parallel specialists)
- LLM orchestrator
- ReAct tool-calling agent
- trained cash-flow forecaster
- Thompson-sampling bandit
- rule fallback (labelled)

Model-agnostic LLM layer — swappable to an India-hosted / Banking-BHASHINI model.

⚙️ Language & runtime

- TypeScript
- Node.js 22+
- native TS execution
- zero npm dependencies
- .env config

Nothing to install; no supply-chain surface.

🌐 Backend

- node:http server
- REST JSON API
- session store
- append-only audit ledger
- retry + timeout client

One process serves the API, the app, and the deck.

🖥️ Front end

- vanilla HTML/CSS/JS
- hand-built SVG charts
- CSS design system
- phone mockup UI
- guided-demo automation

No framework — fast, auditable, demo-proof.

🔒 Compliance rails

- deterministic consent gate
- DPDP-style records
- TRAI window & caps
- human-in-loop routing
- i18n (hi · or · en)

Policy in plain reviewable code, not the model.

📊 Data & measurement

- seeded synthetic generator
- 8 personas
- scoring engine
- 1,500-cohort simulator
- holdout uplift

Zero PII; production swaps in Account Aggregator + YONO events.

~5,700 lines across 33 files · runs with one command: `npm run web`

PROOF — MEASURED ON UPLIFT, NOT CLICKS

Against a randomised **holdout**

+20pp

absolute uplift in positive outcomes · **+334%**
relative

🛡 Stress Shield **+30pp**

👶 New-Baby Nest **+28pp**

📈 First SIP **+24pp**

🕒 Overspend Rescue **+15pp**

26%

PULSE (TREATMENT)

6%

HOLDOUT (BUSINESS AS USUAL)

1,500-customer synthetic cohort, 80/20 split, deterministic — a **labelled simulation** of the measurement methodology that ships with the product. Holdout-based uplift is also the anti-dark-pattern guarantee: we only count **incremental behaviour change**.

THE MOAT

The demo is reproducible. The flywheel isn't.

- 1 Outcome feedback flywheel.** A Thompson-sampling policy learns which tone works per segment from real outcomes — live journeys apply the learned tone and feed results back. **Using the product trains it.**
- 2 Verified learning.** From $\approx$ random start to **+12.9pp over random, 91% of the theoretical ceiling** — and it correctly discovers the hidden best tone for every segment.
- 3 Only compounds at SBI's scale.** Converging the policy needs a stream of consented outcomes only a ~100M-customer base produces. A competitor can clone the UI; they can't clone the data flywheel.
- 4 Compliance lock-in.** Consent, TRAI, audit and human-in-loop wired in from day one — the months-long certification work that blocks every would-be replacement.

Engagement is the **highest-ROI lever** SBI has

15–25%

projected **churn reduction** from proactive engagement

Vantage Point, 2026 retention benchmark

5–7×

cost advantage of retention over acquisition — Pulse works the cheap side

Vantage Point / Backbase, 2026

<₹0.05

LLM cost **per AI conversation turn** — pennies at 100M scale

Computed: ~1.5k in / 300 out tokens × lightweight-LLM list price (\$0.25/\$0.50 per 1M)

0

new dependencies or licences — zero-dep stack, one command to run

Verified in this repo: no npm packages

 **Days, not quarters**

Working product (AI core, 5 surfaces, governance, learning loop) built in **days**. Pilot-ready path: 1–3 months on AA + YONO events.

 **Dormancy → revenue**

Reactivating even a slice of the 34% dormant base and moving products/customer from 2.3 upward is a **board-level number** — with assumptions stated, not overclaimed.

 **Trust dividend**

Care-first + restraint + explainability compounds the asset SBI needs most at 500M scale: **customer trust**.

Phased, measurable, de-risked

Phase 0 · now

Working MVP. Live multi-agent mesh, ReAct banking actions, trained forecaster, guardrails, vernacular, audit trail, uplift measurement, flywheel.

Phase 1 · 1–3 mo

Pilot. Consented Account Aggregator + YONO event streams; real holdout uplift on a volunteer cohort; human-in-loop on all high-value.

Phase 2 · 3–9 mo

Rollout. Push/IVR channels, 10–15 journeys, gamification tie-in with YONO 2.0, Bhashini vernacular, DLT registration.

Phase 3 · 9–24 mo

Platform. Journey Studio for business users, RM copilot, 22 languages, edge/offline, model governance & drift monitoring.

Each phase gates on its own KPIs — engagement lift vs holdout, dormant-revival rate, opt-out rate (low = healthy), products per customer.

Every number on this deck is traceable

FIGURE	SOURCE	NOTE
34% dormant in year 1	StriveCloud, "Banking App Engagement: A Gamification Playbook" (industry data, early 2025)	Global banking-industry figure for new checking accounts; used as a proxy — SBI-specific dormancy would replace it in a pilot.
2.3 products / customer	MatrixBCG, SBI customer demographics & target-market analysis (2025)	SBI's average after CRM-led cross-sell — the baseline Pulse aims to lift.
5–7× retention economics	Vantage Point, "Bank Customer Retention Strategies 2026"; Backbase 2026 retention report	Widely-cited industry benchmark.
~100M YONO users · 20cr target	SBI Chairman C S Setty — PTI via The Week; ZeeBiz (Dec 2025)	9.4–10 crore stated; doubling target within 2 years.
15–25% churn reduction	Vantage Point, 2026	Benchmark for proactive-retention programmes; projected, not measured at SBI.
35% orchestration edge	Gartner estimate, via CMSWire "Agentic AI & Journey Orchestration" (2026)	Operational-efficiency projection for journey-orchestrating banks by 2026.
+20pp / +334% uplift	This build — 1,500-customer synthetic cohort, 80/20 randomised holdout, deterministic	Labelled simulation of the shipped measurement methodology, not field data.
+12.9pp bandit / 91% of ceiling	This build — Thompson-sampling policy, verified run in <code>src/learn/</code>	Real learning algorithm; rewards generated from synthetic ground truth.
<₹0.05 per AI turn	Computed from lightweight-LLM list pricing (\$0.25 in / \$0.50 out per 1M tokens)	~1,500 input + 300 output tokens per reasoning turn ≈ \$0.0005.

Synthetic data throughout (zero PII by design). Projections are labelled as projections; measured numbers come from this repository and are reproducible with one command.



SBI Pulse

- 1 **A multi-agent mesh** — parallel specialists, one orchestrator, a tool-using conversation agent.
- 2 **Knows when not to sell** — stress triggers care; silence is a decision.
- 3 **A compliance cage the AI cannot cross** — the gate lives inside the send tool.
- 4 **Learns from every outcome** — a flywheel that only compounds at SBI's scale.